

Evasion Tactics in Automated Audit and Verification Systems

Abstract

This document provides a comprehensive overview of evasion tactics employed by fraudsters to bypass automated audit and verification software used by auditors, banks, and revenue/financial intelligence authorities. Drawing from recent analyses and industry insights, it expands beyond initial patterns to detail 9 key tactics, including how they exploit limitations in rule-based systems such as static thresholds, predefined patterns, and poor adaptability. Examples from e-commerce, banking, and financial fraud illustrate these tactics, highlighting vulnerabilities in tools like ACL Analytics and IDEA (for auditors), Actimize and Oracle Financial Services (for banks), and goAML or IRS Electronic Fraud Detection System (EFDS) (for authorities). The analysis emphasizes the need for general-purpose AI to counter these adaptive threats. APA-style citations reference empirical sources on fraud evasion and detection advancements.

Introduction

Automated audit and verification software is essential for financial integrity, yet fraudsters continually adapt to evade detection. Rule-based systems, which dominate tools used by auditors (e.g., ACL for data querying and anomaly flagging), banks (e.g., Actimize for real-time transaction monitoring), and authorities (e.g., goAML for suspicious activity reporting), rely on predefined rules, thresholds, and patterns. These create exploitable gaps, such as inability to handle novel variations or unstructured data. This document details 9 evasion tactics, focusing on their mechanisms against specific systems, with examples from financial contexts.

Evasion Tactics

Fraudsters engineer activities to circumvent rule-based algorithms by exploiting static nature, high false positives, and limited real-time adaptability. Below are 9 key tactics, each with mechanisms, examples, and evasion specifics.

1. Simulation of Legitimacy Fraud is structured to mimic statistical norms, such as bell-

shaped distributions, avoiding outliers that trigger alerts. For instance, small repeated debits (e.g., 651 INR on 2021-09-12) appear as random errors. In auditors' tools like ACL, which use statistical sampling for variance checks, this blends with legitimate data, evading deviation-based flags. [datadome.co](#) Banks' Actimize employs rules for unusual patterns, but mimicry reduces alerts by aligning with customer baselines. [tookitaki.com](#) For authorities like IRS EFDS, which scans for structuring below thresholds, simulated normalcy fails to trigger profiling-based SARs. [sanctions.io](#) This exploits systems' reliance on predefined indicators (Ngai et al., 2011).

2. **Gradual Escalation** Start with minor probes (e.g., zero-value payments like -100 INR on 2022-04-05) and scale undetected, evading spike-focused detection. Auditors' IDEA trend analysis prioritizes abrupt changes, allowing incremental overcharges (e.g., to 32 lakh INR) to pass as organic. [digitaldefynd.com](#) In banking Oracle Mantas, velocity rules (e.g., >X transactions/day) miss slow ramps mimicking growth. [dl.gi.de](#) Authorities' goAML chronological scans overlook trajectories below immediate limits, enabling undetected layering (Phua et al., 2010). [complyadvantage.com](#)
3. **Batch and Offset Tactics** Use paired positives/negatives (e.g., +141,800/-141,600 INR on 2022-04-05) to balance aggregates. In auditing TeamMate, aggregate checks see balanced ledgers, concealing net gains in batches. [fraud.com](#) Banks' payment netting in Actimize views pairs as legitimate, evading unmatched alerts. [kount.com](#) FIUs' threshold reporting (e.g., CTRs >\$10,000) treats balanced batches as non-suspicious (Albrecht et al., 2015). [tookitaki.com](#)
4. **Threshold Avoidance** Keep activities below alert levels, e.g., splitting large sums into small transactions to evade value-based flags. Auditors' ACL threshold queries (e.g., >\$500) miss distributed fraud like split e-commerce purchases. [seon.io](#) Banks' Oracle uses amount rules; fraudsters stay under to avoid AML triggers in smurfing. [sciencedirect.com](#) IRS EFDS structuring detection fails when splits mimic normal small transfers. [fraud.net](#) This circumvents static limits (Hilal et al., 2022).

5. **Velocity and Timing Manipulation** Space actions or use off-hours to avoid frequency

checks, e.g., limiting logins to <5 in 5 minutes. IDEA's pattern-matching in audits misses spaced multiples over days. [seon.io](#) Actimize velocity rules (e.g., rapid spending) are evaded by gradual pacing in banking. [seon.io](#) goAML's time-based filters overlook off-peak laundering. [cockroachlabs.com](#) Exploits fixed time windows.

6. Data Obfuscation Omit or falsify data (e.g., missing IDs in 100% of OTP bypass records), hindering traceability. ACL's data validation struggles with unstructured inputs, allowing phantom payments. [protechtgroup.com](#) Oracle's profiling fails without complete fields in banking. [fraud.com](#) IRS EFDS can't detect concealment in incomplete reports. [netnow.io](#) Leverages data limitations (PMC, 2023).
7. Cross-Channel Exploitation Use silos across products (e.g., banking to wallets), evading single-channel monitoring. Auditors' IDEA misses inter-product links in aggregated data. [tookitaki.com](#) Actimize may not correlate across channels, allowing ATO via phishing then transfers. [tookitaki.com](#) goAML's reporting silos permit cross-border layering. [tookitaki.com](#) Exploits fragmented systems.
8. Noise Injection Generate false positives via high-frequency low-impact actions, causing alert fatigue. ACL's anomaly flagging overwhelms auditors with noise. [f5.com](#) Oracle's real-time alerts in banks lead to ignored high-impact fraud. [fraud.com](#) IRS EFDS SAR reviews miss amid volume. [f5.com](#) Diverts attention.
9. Adversarial Adaptation Use AI to mimic legitimate behavior or create deepfakes, evolving faster than rules. IDEA's static patterns can't counter AI-generated variations. [datadome.co](#) Actimize ML components are fooled by adversarial inputs in banking phishing. [f5.com](#) goAML's pattern recognition misses adaptive laundering. [f5.com](#) Outpaces updates.

Flaws in Automated Systems

These tactics exploit key flaws: inability to adapt (miss evolving schemes), high false positives (reduce trust), and data limitations (fail unstructured inputs), with rule-based detection at 60–70% efficacy (ResearchGate, 2024).

The Need for General-Purpose AI

AI addresses these by learning dynamically, e.g., detecting novel patterns with 15–30%

better accuracy (ResearchGate, 2024b).

Conclusion

These 9 tactics sustain fraud by exploiting rule-based rigidity. AI adoption is urgent for adaptive detection.

References

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